

Andy Anderson

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PROFESSIONAL SUMMARY

Software engineer specializing in secure backend development, with experience designing, building, and maintaining dynamic, data-driven applications. Skilled in the full software development life cycle, from analyzing business needs to implementation, testing, and post-deployment support. Strong knowledge of microservices, relational databases, cloud deployment, and skilled in working across collaborative agile development environments.

SKILLS

- **Languages:** Python, JavaScript, C#, Java, SQL, HTML/CSS
- **Frameworks & Architecture:** .NET Core, ASP.NET, REST APIs, Microservices, Vue.js, React.JS, AngularJS
- **Tools & Platforms:** AWS, Azure, MS SQL Server, MySQL, Visual Studio, Git, Jenkins

EDUCATION

Western Governors University

B.S. in Computer Science – 3.97 GPA

Oct 2023

Salt Lake City, UT

WORK EXPERIENCE

Backend Engineer

Microchip Technology

Oct 2025 – Present

Phoenix, AZ

- Leading system architecture and backend development of a real-time semiconductor inventory, pricing, and sourcing platform for authorized distributors, projected to support over \$30M in annual transaction volume.
- Identified and remediated critical client-side field manipulation and spoofing vulnerabilities in checkout flow by implementing stricter server-side validation and data integrity controls to prevent fraud and inventory loss.
- Integrating Retrieval-Augmented Generation (RAG) into internal LLMs to deliver contextual, data-driven answers from product datasheets and documentation, accelerating engineering design and development workflows.

Software Security Engineer

Marcus Engineering

Jan 2024 – Oct 2025

Tucson, AZ

- Designed and developed a distributed, multi-tiered GPS tracking and IoT web platform for portable elevator lifts using C# and .NET Core, building responsive dashboards with real-time telemetry for a \$600k contract.
- Identified and reported security vulnerabilities in the REST API endpoints for an industry-leading semiconductor company's conference web application while in attendance, earning executive recognition.
- Designed relational databases in MS SQL Server and secure key management in Azure for medical instrument reprocessing applications, enabling high-volume ingestion with 50% faster queries than previous implementations.
- Developed a desktop application for USB specification testing and validation, including power delivery and transfer speed; licensed for use in certifying military-adjacent mobile products, reducing time-to-market by six months.
- Contributed to adoption of Git flow standards, CI/CD pipelines in Jenkins, and standardized testing frameworks, helping reduce deployment times and improve code quality across projects.
- Facilitated requirements gathering, algorithm design, solution implementation, and post-deployment customer support for consulting clients, helping secure repeat contracts and long-term relationships with 4 large companies.
- Mentored interns through code reviews and design pattern guidance, reducing onboarding time and improving collaboration and code quality across the team.

Undergraduate Researcher (Software)

Seedscape Ecology Laboratory

Sep 2020 – May 2021

Logan, UT

- Developed analysis software in Python and scripts in R to evaluate collected biochemical data.
- Contributed to research leading to \$1M NSF grant and scientific journal publication.

PERSONAL PROJECTS

- **BrickPlanet Sniper** – Built a virtual collectible item purchase bot using RESTful APIs and HTTP requests; scaled to 30+ recurring paid users as a monetized SaaS.
- **Everything Toolbar** – Contributed to an open-source C# desktop file search tool with over 12k stars on GitHub.
- **To The Moon** – Developed an Android mobile game where players physically throw their phones in the air for points.